python tasks :-

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1. Shopping Bill

A customer buys:

Rice = ₹500

Oil = ₹200

Sugar = ₹150

Calculate:

Total amount

GST @ 18%

Final bill amount

2. Mobile Purchase

A mobile phone costs ₹25,000.

Calculate:

10% discount

Discount amount

Final payable amount

3. Employee Salary

An employee's basic salary is ₹30,000.

Calculate:

HRA = 20%

DA = 10%

Total salary

4. Student Marks

Marks:

English = 80

Maths = 90

Science = 85

Calculate:

Total marks

Average marks

Percentage

5. Restaurant Bill

A family orders:

Biryani = ₹300

Cool Drinks = ₹100

Ice Cream = ₹150

Calculate:

Total bill

GST @ 5%

Final bill

6. Electricity Bill

Units consumed = 250

Rate per unit = ₹8

Calculate:

Total bill amount

7. Movie Ticket Booking

Ticket price = ₹200

Number of tickets = 5

Calculate:

Total cost

GST @ 12%

Final amount

8. Online Course Fee

Course fee = ₹15,000

Discount = 20%

Calculate:

Discount amount

Final fee

9. Bank Fixed Deposit

Principal = ₹50,000

Rate = 6%

Time = 2 years

Calculate:

Simple Interest

Total Amount

10. Petrol Expense

Petrol price per litre = ₹105

Litres purchased = 12

Calculate:

Total petrol cost

11. Laptop Purchase

Laptop price = ₹60,000

Discount = 15%

GST = 18%

Calculate:

Discount amount

Price after discount

Final price after GST

12. House Rent Calculation

Monthly rent = ₹12,000

Calculate:

Rent for 6 months

Rent for 1 year

13. Cricket Score

Runs scored in 5 matches:

45, 67, 89, 34, 78

Calculate:

Total runs

Average runs

14. Grocery Store

Items:

Milk = ₹60

Bread = ₹40

Eggs = ₹80

Calculate:

Total bill

GST @ 5%

Final amount

15. Salary Increment

Current salary = ₹40,000

Increment = 10%

Calculate:

Increment amount

New salary

16. Bike EMI

Bike price = ₹1,20,000

Down payment = ₹20,000

Calculate:

Loan amount remaining

17. School Fee

Tuition Fee = ₹25,000

Exam Fee = ₹2,000

Bus Fee = ₹5,000

Calculate:

Total annual fee

18. Water Consumption

Water consumed = 15,000 litres

Cost per litre = ₹0.02

Calculate:

Total water bill

19. Data Analyst Salary Package

Basic Salary = ₹50,000

Bonus = ₹10,000

Calculate:

Annual salary package

20. E-commerce Order

Product Price = ₹2,500

Shipping Charge = ₹100

GST = 18%

Calculate:

Total amount payable

Easy Level

Task 1

Process a list of student names and print:

Plain text

Processing Student: Name

Task 2

Process a list of employee names and print attendance status.

Task 3

Print only Gmail accounts from a list of emails.

Task 4

Convert all names in a list to uppercase.

Task 5

Remove extra spaces from employee names before processing.

Task 6

Count the total number of products in a shopping cart.

Task 7

Calculate the sum of all salaries in a salary list.

Task 8

Print only even numbers from a list.

Task 9

Print only odd numbers from a list.

Task 10

Find the highest mark from a list of student marks.

Medium Level

Task 11

Stop processing when an empty student name is found.

Task 12

Skip weekend days while processing calendar days.

Task 13

Validate mobile numbers and print valid/invalid.

Task 14

Print employees whose salary is greater than ₹50,000.

Task 15

Find all failed students (marks below 35).

Task 16

Calculate bonus for employees based on salary.

Task 17

Process order IDs and stop if a cancelled order is found.

Task 18

Find duplicate values in a list.

Task 19

Count vowels in each employee name.

Task 20

Print products whose stock is below 10.

String Processing Tasks

Task 21

Convert all file names to lowercase.

Task 22

Convert all .txt files to .csv.

Task 23

Print usernames that start with "admin".

Task 24

Print emails ending with .com.

Task 25

Find usernames containing special characters.

Task 26

Detect SQL injection keywords like:

Plain text

DROP

DELETE

TRUNCATE

Task 27

Print names having more than 5 characters.

Task 28

Replace spaces in file names with underscores.

Task 29

Count how many Gmail accounts exist.

Task 30

Print only unique names from a list.

Task 1 – Student Management System

Scenario

A school wants to maintain student results.

Requirements

Ask the user to enter the student name.

Ask the user to enter marks for 5 subjects using a loop.

Calculate the total marks.

Calculate the average marks.

Display:

Pass if average is 35 or above.

Fail if average is below 35.

Assign grade:

A → Average 90 and above

B → Average 75 to 89

C → Average 60 to 74

D → Average 35 to 59

F → Below 35

Ask the user whether they want to enter another student.

Continue until the user enters "No".

Display:

Total students processed

Total passed students

Total failed students

Task 2 – ATM PIN Validation System

Scenario

An ATM allows only 3 PIN attempts.

Requirements

Store the correct PIN as 1234.

Ask the user to enter the PIN.

If PIN is correct:

Display "Login Successful".

Display account balance.

If PIN is incorrect:

Display "Wrong PIN".

Allow only 3 attempts using a loop.

If all 3 attempts fail:

Display "Account Blocked".

Count and display total attempts used.

Task

GroupBy Analysis

1. Find average Marks, max Marks, and student count per Subject

2. Find the pass rate (Marks ≥ 50) per City

3. Add a column showing each student's deviation from their subject average

4. Create a pivot table: Subject × Grade showing count of students

5. Filter to keep only Cities where average Marks > 75

Task 1: Number Pyramid

Problem: Write a program to print a pyramid of numbers up to n.

Task 2: Alphabet Triangle

Problem: Print a triangle of alphabets starting from A.

Task 3:Hourglass pattern

Combination of inverted and upright pyramids.

Tasks

Write a Python program to find the most repeated word in a given paragraph.

Write a Python program to find the least repeated word in a given paragraph.

Write a Python program to remove duplicate words from a paragraph while maintaining the original order of words.

Write a Python program to count the number of uppercase letters, lowercase letters, digits, and special characters present in a paragraph.

Write a Python program to reverse the complete paragraph.

Write a Python program to reverse each word in a paragraph without changing the order of the words.

Today's Tasks

Write Python programs for the following:

1. Palindrome Number

Take input from the user.

Check whether the given number is a palindrome or not.

Display an appropriate message.

Example:

Input: 121

Output: 121 is a Palindrome Number

2. Perfect Number

Take input from the user.

Find the sum of all proper factors (excluding the number itself).

Check whether the number is a Perfect Number.

Display an appropriate message.

Example:

Input: 28

Output: 28 is a Perfect Number

3. Strong Number

Take input from the user.

Calculate the factorial of each digit.

Find the sum of the factorials of all digits.

Check whether the number is a Strong Number.

Display an appropriate message.

Example:

Input: 145

Output: 145 is a Strong Number

Challenge Task

Design a Movie Ticket Booking function:

def book_ticket(customer_name, movie_name, tickets=1, *snacks, **details):

Requirements:

Print customer name, movie name, and number of tickets.

Print all snacks ordered.

Print theatre, seat type, payment mode, and show time.

Calculate the total ticket cost.

Challenge Task

Bank Loan EMI

def loan_details(customer_name, loan_amount, years=1, *charges, **details):

Requirements

Print customer name.

Print loan amount.

Print processing charges.

Print interest rate.

Calculate EMI.

Print total repayment.

Hospital Bill

def hospital_bill(patient_name, disease, days=1, *tests, **details):

Requirements

Print patient details.

Print all medical tests.

Print doctor name and room type.

Calculate room charges.

Add test charges.

Print final hospital bill.

Scenario-Based Exercises1. Student Marks (Tuples)Input: [("Alice", 85), ("Bob", 35), ("Charlie", 90)]Use filter → Select students who passed (marks ≥ 40)Use sorted → Rank students by marksUse map → Add 5 grace marks to everyoneUse reduce → Find total class score

2. Shopping Cart (Prices)Input: [1200, 850, 600, 1500]Use map → Apply 10% discountUse filter → Keep only items above ₹1000Use reduce → Calculate total billUse sorted → Sort prices ascending

3. Bank TransactionsInput: [5000, -1200, 3000, -700, 1500]Use filter → Extract only debits (negative values)Use map → Convert all amounts to USDUse reduce → Find net balanceUse sorted → Sort transactions by amount

4. Employee RecordsInput: [("Alice", 50000), ("Bob", 45000), ("Charlie", 60000)]Use sorted → Sort employees by salaryUse map → Increase salary by 10%Use filter → Select employees earning ≥ 50,000Use reduce → Find total payroll

Tasks

Online Food Order

def food_order(customer, amount):

Inner functions:

Discount

Delivery Charge

GST

Final Amount

Hospital Bill

def hospital(patient_name, room_charge, medicine_charge):

Create inner functions:

Room Bill

Medicine Bill

Total Bill

ATM Machine

def atm(balance):

Create inner functions:

deposit(amount)

withdraw(amount)

check_balance()

Task 1

Validate Email

Conditions

Username

@

Domain

.com/.in/.org

task 2

Username Validation

Conditions

Starts with alphabet

Minimum 5 characters

Maximum 15

Only letters, numbers and underscore

task 3

PAN card validation

Assignment 1 : Find Total Days Lived

Problem Statement

Write a Python program that accepts the user's Date of Birth.

Calculate the total number of days the user has lived from their birth date until today.

Sample Input

Enter Date of Birth: 01-08-1998

Expected Output

You have lived 10199 days.

Assignment 2 : Find Next Birthday and Remaining Days

Problem Statement

Write a Python program that accepts the user's Date of Birth.

Your program should:

Find the user's next birthday.

Calculate how many days are left until that birthday.

If the birthday has already passed this year, calculate it for the next year.

Sample Input

Enter Date of Birth: 01-08-1998

Expected Output

Next Birthday: 01-08-2026

Days Remaining: 23

Assignment 3 : Calculate Employee Experience

Problem Statement

Write a Python program that asks the employee to enter their Joining Date in DD-MM-YYYY format.

Calculate the employee's total experience in years.

Sample Input

Enter Joining Date: 15-06-2022

Expected Output

Experience: 4 Years

Assignment 4 : Employee Late Login Checker

Problem Statement

Assume office timing is 09:30 AM.

Write a Python program that accepts an employee's login time in HH:MM format.

The program should:

Check whether the employee logged in late.

If late, display the late duration.

Otherwise, display "Logged in on time."

Sample Input

Enter Login Time: 09:45

Expected Output

Status: Late Login

Late By: 0:15:00

Task

Create an Employee class with attributes:

Employee ID

Employee Name

Department

Salary Display the employee details.

Use a class variable to store the company name shared by all employees.

Count the total number of employees using a class variable.

Create a class method to change the company name for all employees.

Create a static method to calculate income tax based on salary.

Create a static method to validate an employee ID format.

Make the salary attribute private and provide getter and setter methods.

Validate the salary using a setter (salary cannot be negative).

Delete an employee's salary using a property deleter.

Create a read-only Employee ID using the @property decorator.

Prevent users from assigning a salary greater than ₹10,00,000.

Library Management System

Concepts: Class interactions, full system design.

Task: Build three interacting classes:

Before writing code, map out the 3 classes using this framework:

1. Book (The Data Container)

Think of a physical book sitting on a library shelf. It doesn't "do" much on its own, but it holds important information.

Attributes (What it HAS):

title (string): e.g., "The Hobbit"

author (string): e.g., "J.R.R. Tolkien"

isbn (string): A unique ID barcode for every book (e.g., "978-0547928227").

is_checked_out (boolean): Tracks availability (True or False).

Why start is_checked_out as False?

When a library buys a brand-new book, it starts on the shelf ready for borrowing!

Methods (What it DOES):

__init__(): The constructor method that assigns the attributes when a new book is created.

__str__(): A special helper method to return a readable text string describing the book (e.g., "The Hobbit by J.R.R. Tolkien [Available]").

2. Member (The Actor)

Think of a person holding a library card. They interact with books and have rules to follow.

Attributes (What it HAS):

name (string): e.g., "Alice"

member_id (string): Unique ID card number (e.g., "M001").

borrowed_books (list): A list that stores actual Book objects the member currently has in their backpack.

Key Concept to Explain Students:

borrowed_books should store Book objects, not just titles as text! Storing objects lets the member access properties like book.author or book.isbn.

Methods & Rules (What it DOES):

borrow_book(book): Adds a book object to borrowed_books.

return_book(book): Removes a book object from borrowed_books.

can_borrow(): A helper method that checks if len(borrowed_books) < 3.

3. Library (The Manager / System)

The Library is the central controller. It doesn't represent a person or a single item; it represents the system that coordinates everything.

Attributes (What it HAS):

books (dictionary or list): Holds all books owned by the library. (Tip: Using a dictionary where the key is the isbn makes looking up books super easy!)

members (dictionary or list): Holds all registered members. (Tip: Using a dictionary with member_id as keys makes user lookup effortless!)

Methods (How it coordinates):

add_book(book): Adds a new Book object to the library's collection.

register_member(member): Adds a new Member object to the library's database.

checkout_book(member_id, isbn): The most important method! It handles validation and connects a Member to a Book.

return_book(member_id, isbn): Reverses the checkout process.